

Abstract

A topological interlocking assembly (TIA) consists of multiple blocks, which are in contact with each other, such that there exists a subset of the blocks which if fixed in place, already fixes all other blocks through contact. Geometries admitting a TIA are known since 1735, however until recently the number of them was very low. Recent mathematical advances present a new method of constructing large numbers of candidate geometries, which in all investigated cases admit a TIA. The method is based on the interpolation of deformed fundamental domains of crystallographic groups.

This talk will present the new method as well as a generalisation to the three dimensional case. Additionally, we present results about the mechanical performance of these newly generated TIAs and new approaches to more efficiently evaluate them. Further, we present applications of TIAs from multiple interdisciplinary collaborations, including aerospace, medical devices and biology connected to civil engineering.

As this is ongoing work it might contain preliminary results and present current research questions.

This is joint work with Alice C. Niemeyer and Kai-Uwe Schröder.

Literature